



# NEW PRODUCTS

## COMPONENTS

### Microcontrollers

Renesas Electronics has announced the expansion of its RX24T and RX24U groups of 32-bit microcontrollers (MCUs) to include new high-temperature-tolerant models for motor-control applications that require an expanded operating



temperature range. The new RX24T G and RX24U G versions support operating temperatures ranging from -40°C to +105°C, while maintaining high speed, high functionality and energy efficiency of RX24T and RX24U MCUs.

Renesas Electronics Singapore Pte Ltd  
[www.renesas.com](http://www.renesas.com)

### Motion sensor

STMicroelectronics has integrated machine learning technology into its advanced inertial sensors to improve activity-tracking performance and battery life in mobiles and wearables. The LSM6DSOX iNEMO sensors contains a machine learning core to classify motion data based on known patterns. Relieving this first stage of activity tracking from the main processor saves energy and accelerates motion-based apps such as fitness logging, wellness monitoring, personal navigation and fall detection.

STMicroelectronics  
[www.st.com](http://www.st.com)



### Smart BLDC motor

PD42-x-1670 is an easy-to-use and compact PANdrive smart brushless DC (BLDC) motor. The module is controlled via CAN bus interface and comes with two firmware options, namely, TMCL and

CANopen. The PD42-x-1670 line offers an integrated hall-sensor-based encoder for closed-loop FOC-based operation. It enables machine builders to implement rapid, precise and efficient positioning in no time, making it suitable for lab automation, factory automation, manufacturing, robotics and CNC machines.

Trinamic  
[www.trinamic.com](http://www.trinamic.com)

### MOSFETs

Addressing the fast-growing demand for silicon-carbide (SiC) solutions, Infineon Technologies has launched new devices in the 1200V CoolSiC metal oxide semiconductor field effect transistor (MOSFET) family. CoolSiC Easy 2B power modules enable engineers to reduce system costs by increasing power density. In addition, these can also lower operational costs significantly. Owing to about 80 per cent lower switching losses compared to silicon insulated-gate bipolar transistors (IGBTs), inverter efficiency levels exceeding 99 per cent can be reached. Because of specific SiC properties, the same or even higher switching-frequency operation can be realised. This is particularly attractive for fast-switching applications such as UPSes and energy storage.

Infineon Technologies India Pvt Ltd  
[www.infineon.com](http://www.infineon.com)

### BLDC fan drivers

Allegro MicroSystems has launched QuietMotion family consisting of the first-to-market field-oriented control BLDC electric motor controllers that are customer code-free. The devices are designed to provide reliable and efficient



low audible noise performance while reducing design cycle time.

With its embedded proprietary windowless sinusoidal algorithm and soft-on-soft-start algorithms, AMT49406 provides reliable, industry-leading audible noise performance.

A89301 offers ultra-quiet operation, as well as additional flexibility and programmable features for premium appliances. It is specifically targeted at systems with low speed requirements.

Allegro MicroSystems  
[www.allegromicro.com](http://www.allegromicro.com)

### DC/DC converters

REC15E-Z DC/DC converters are fully-specified devices with 15W, no minimum load, 1600V DC isolation, high efficiency up to 90 per cent and low ripple/noise. The REC15E-Z series was designed for cost-sensitive applications where board space is at a premium. The wide 4:1 input ranges accept 9V-36V or 18V-75V to cover multiple supply options such as lead-acid or lithium batteries or 12V/24V/36V/48V industrial bus voltages. Inputs are protected against transients of up to 100V and feature under voltage-lockout to protect batteries from being over-discharged. Single or dual outputs are continuously protected against short-circuit and overload conditions, and can drive high-capacitive loads. These are fully certified to industrial EMC and safety standards, and come with a three-year warranty.

RECOM Power  
[www.recom-power.com](http://www.recom-power.com)



### Pressure switches

Falcon Engineering has launched pressure switches that monitor the pressure of water media and close or open an electrical circuit on reaching set threshold. This threshold level can be set manually using M4 grub screw provided on the switches. There are two